



LGL2: Pathology of Esophagus

By: Consultant Pathologist

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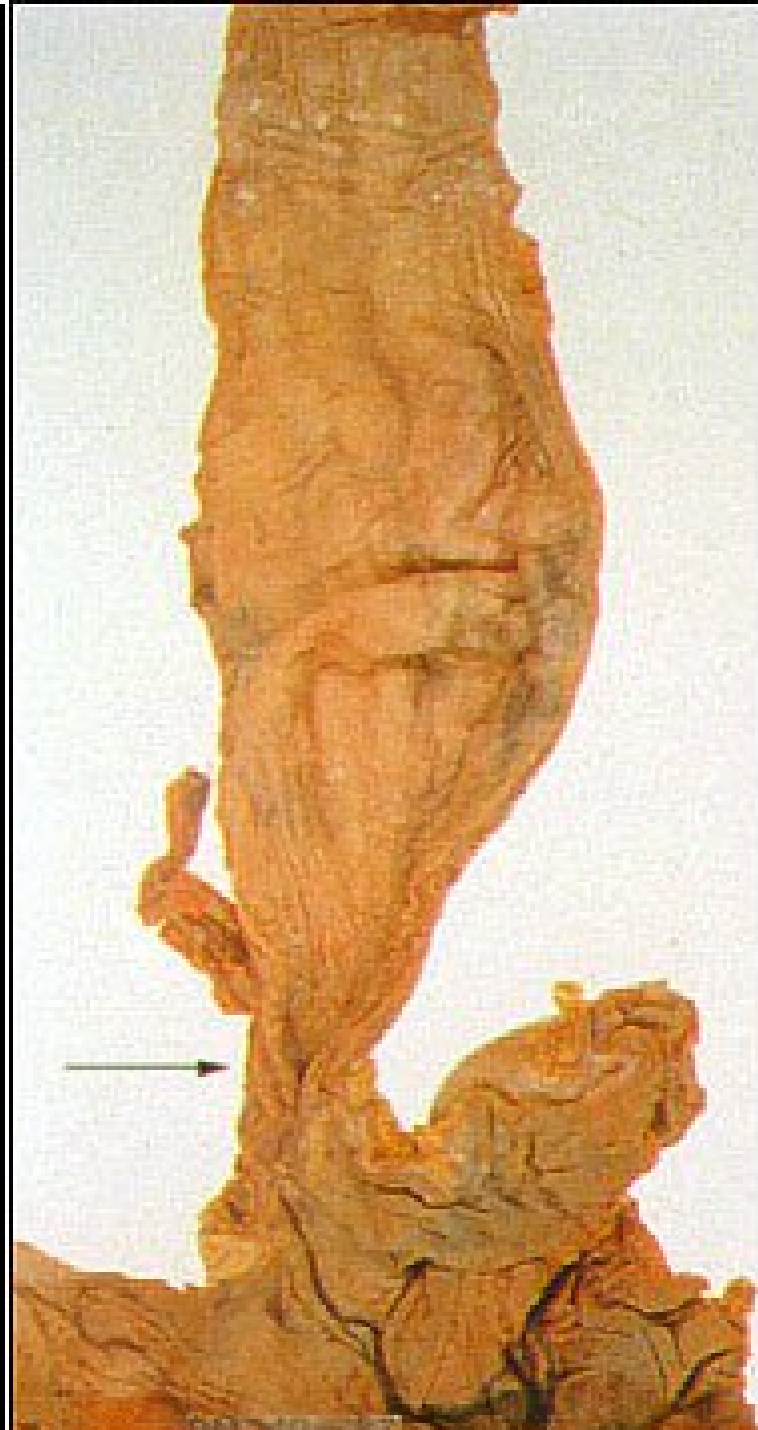
Obstructive and Vascular diseases

Achalasia:

- Motility disorder of esophagus caused by increased lower esophageal sphincter (LES) tone due to absence of myenteric plexus in the body of esophagus with consequent functional esophageal obstruction.
- Characterized by the triad of:
 - Increased LES tone
 - Incomplete LES relaxation
 - Esophageal aperistalsis

- Types of Achalasia:
 - Primary (idiopathic) : caused by failure of distal esophageal inhibitory neurons.
 - Secondary: as in Chagas disease (Trypanosoma Cruzi infection) which causes destruction of myenteric nervous plexus.
- **Achalasia-like disease (Pseudo Achalasia):** caused by
 - Diabetic autonomic neuropathy.
 - Infiltrative disorders: such as malignancy, amyloidosis, or sarcoidosis.
 - Lesions of dorsal motor nuclei, which may be produced by polio or surgical ablation.

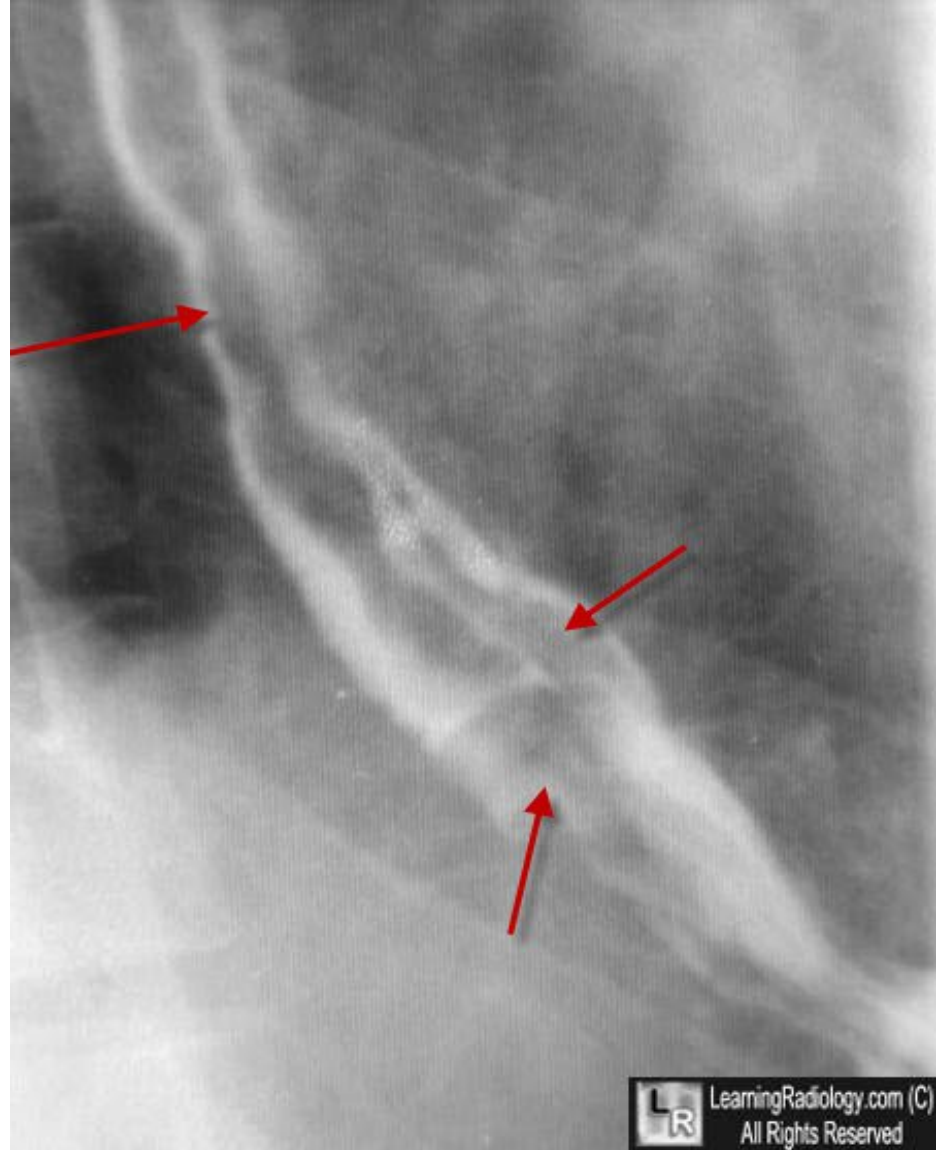
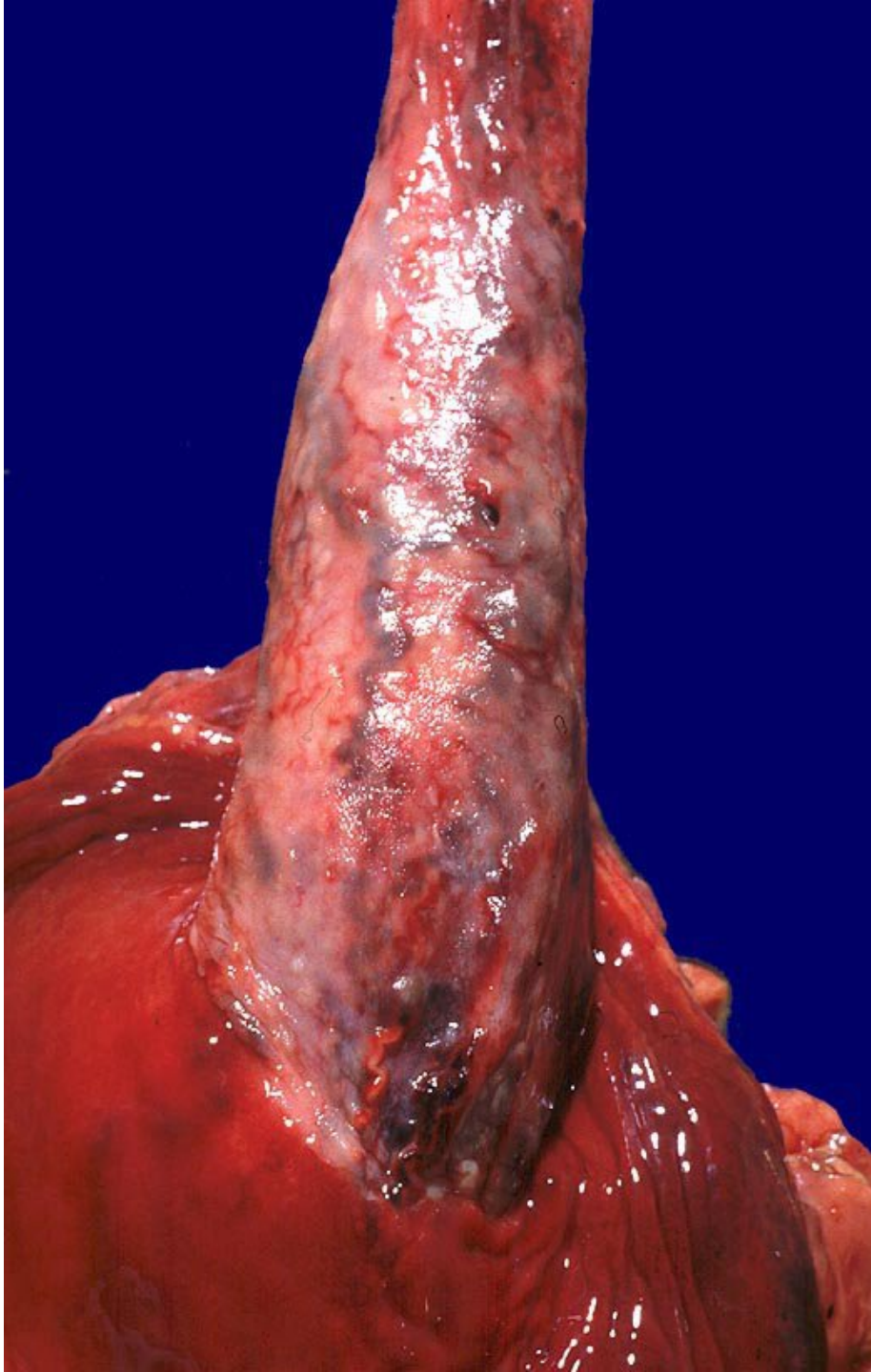
Achalasia



Esophageal Varices:

- Dilated veins in submucosal venous plexus within the distal esophagus that are prone to rupture and hemorrhage.
- Causes:
 - Liver cirrhosis and portal hypertension (>90%), mostly alcohol induced.
 - Hepatic schistosomiasis

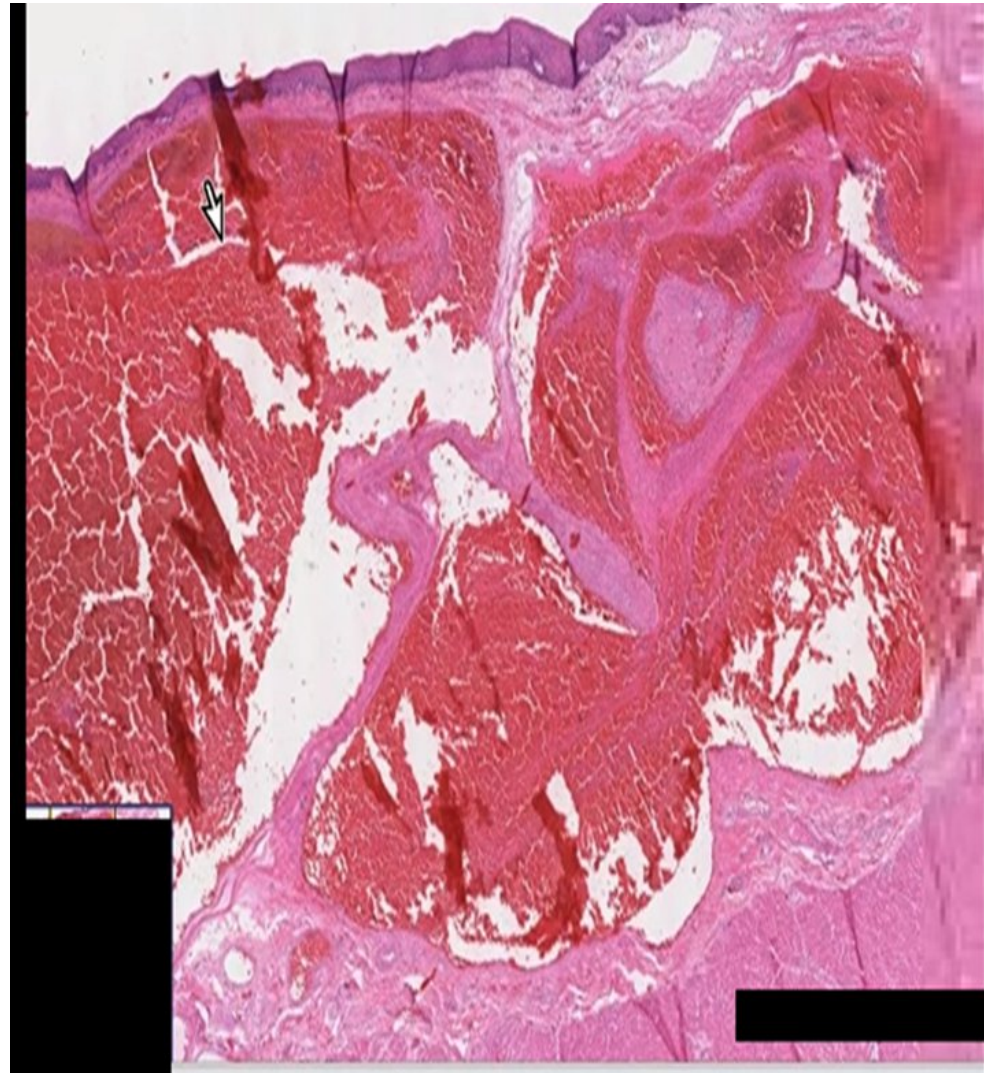
- The upper gastric & lower esophagus veins are linked to portal system via gastroesophageal anastomoses.
- In portal hypertension these anastomoses become prominent, dilate & become varicose.
- If varices exceed 5 mm in diameter, they may rupture, leading to life-threatening bleeding.



The lower esophagus during a barium swallow shows multiple serpiginous filling defects in the barium column (**red arrows**) from varices protruding into the lumen from the wall.



Figure 1 Large esophageal varices at EGD.



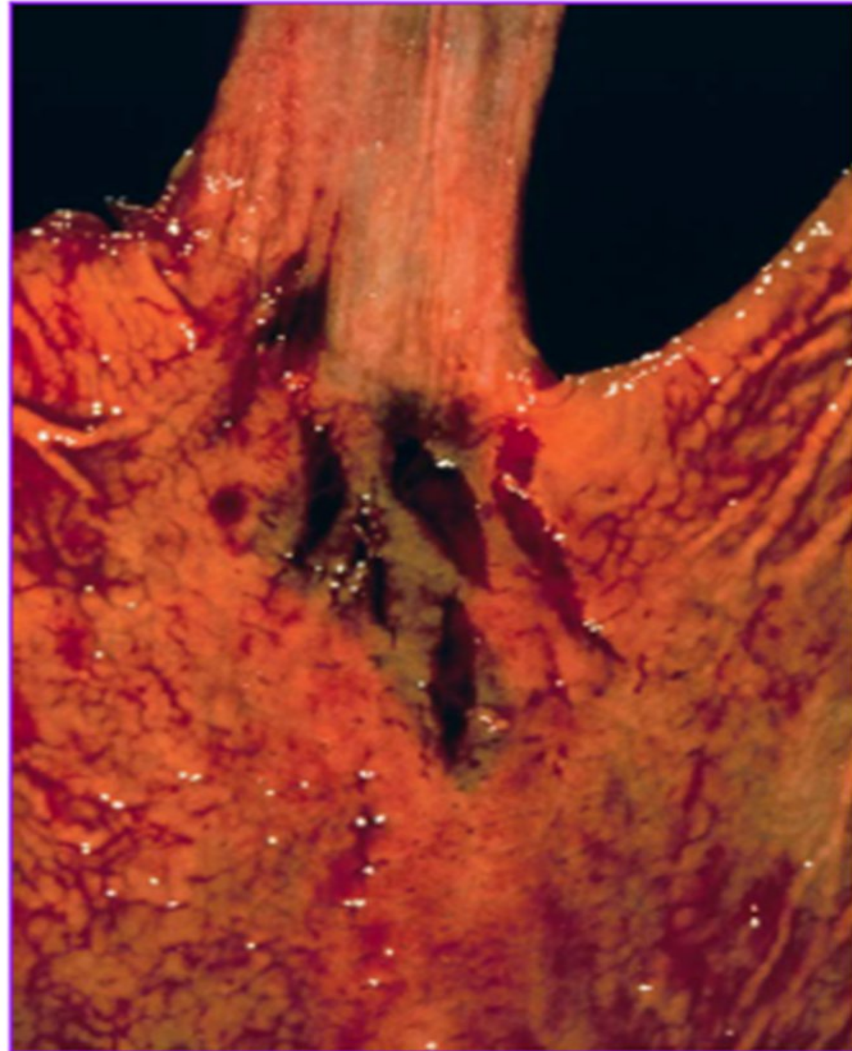
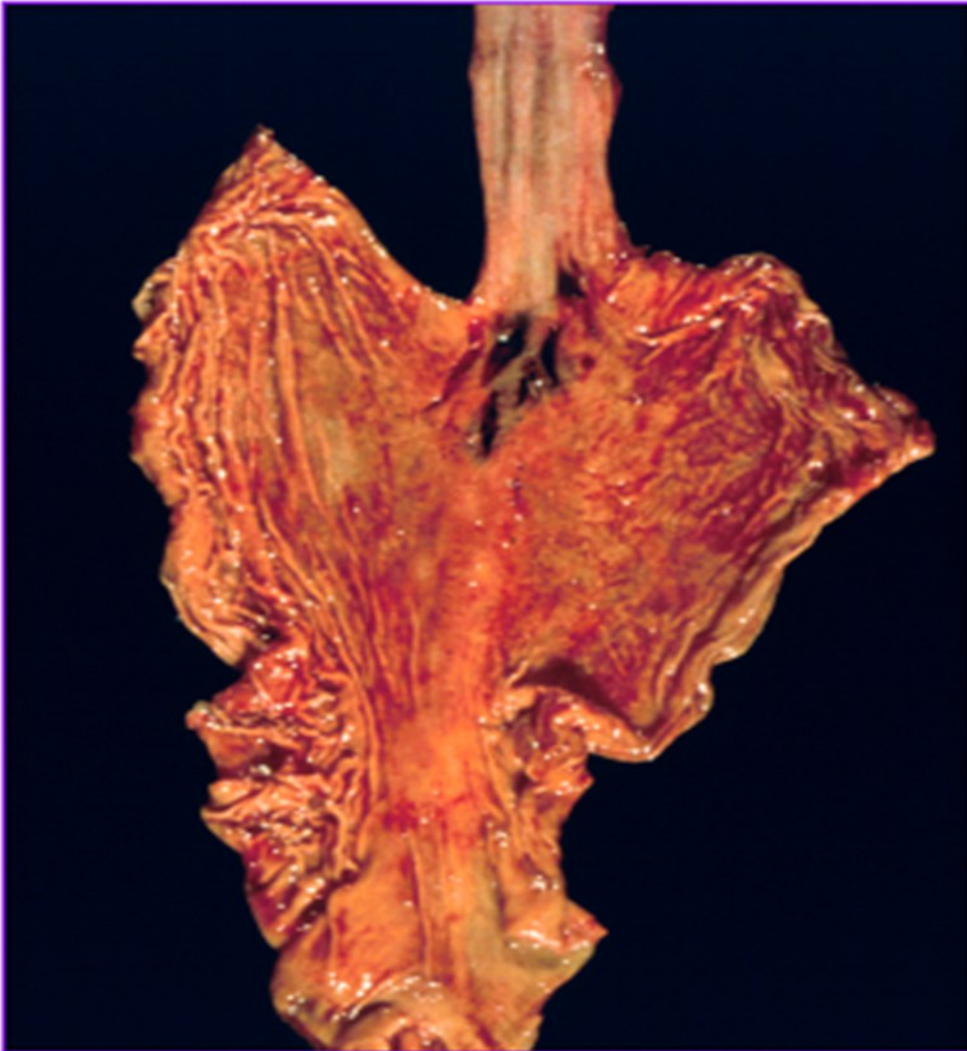
The overlying mucosa can be intact (as seen here) but is ulcerated and necrotic if rupture has occurred.

Esophageal Lacerations

Mallory-Weiss tears:

- Longitudinal **mucosal** tears near GE junction.
- Often associated with severe retching and vomiting secondary to acute alcohol intoxication.
- It's due to failure of reflex relaxation of GE musculature that precedes the contractile wave.

- The tears are longitudinally oriented and range in length from millimeters to several centimeters.
- They present with GI bleeding, hematemesis.



Boerhaave syndrome:

- A **transmural** tearing and rupture of distal esophagus
- The factors giving rise to this syndrome are similar to those for Mallory-Weiss tears but more severe.
- This produces severe mediastinitis.
- Patients present with severe pain, tachypnea, and shock.
- Usually fatal.



Esophagitis (Inflammation of Esophagus)

Causes & Types:

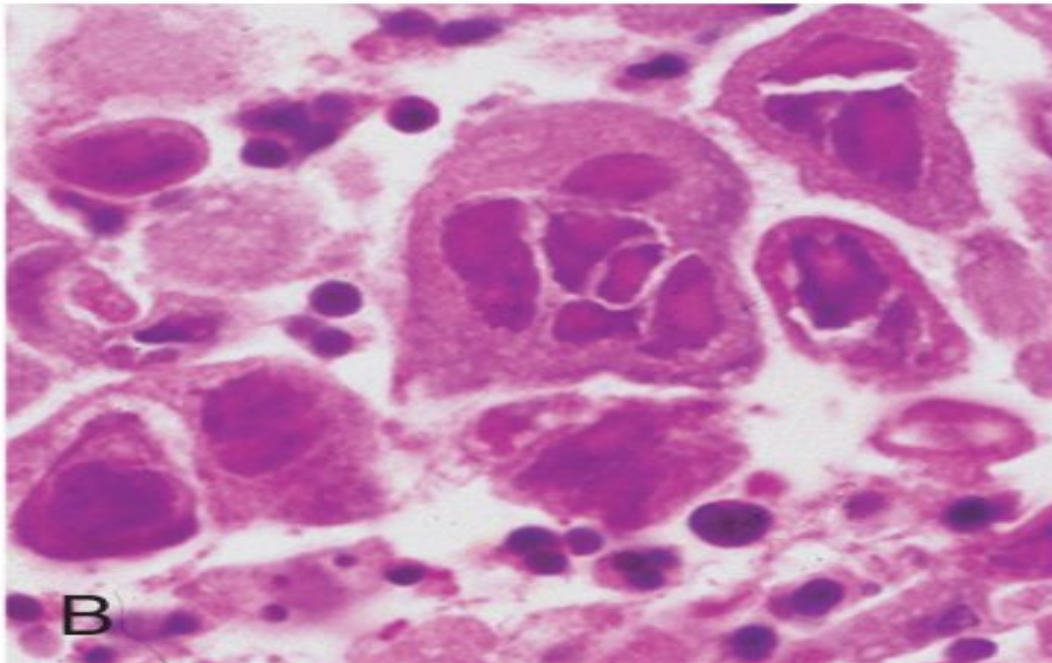
- **Reflux Esophagitis**
- **Chemical: alcohol, erosive acids & alkali.**
- **Medication: pill-induced esophagitis, chemotherapy.**
- **Infectious esophagitis**
- **Eosinophilic Esophagitis**
- **Radiation therapy.**
- **Desquamative skin diseases may involve esophagus e.g. bullous pemphigoid and epidermolysis bullosa.**
- **Crohn disease.**
- **Graft-versus-host disease.**

Infective Esophagitis

- Most frequent in debilitated or immunosuppressed patients.
- Include:
 - Viral: e.g. HSV & CMV
 - Fungal: e.g. Candida

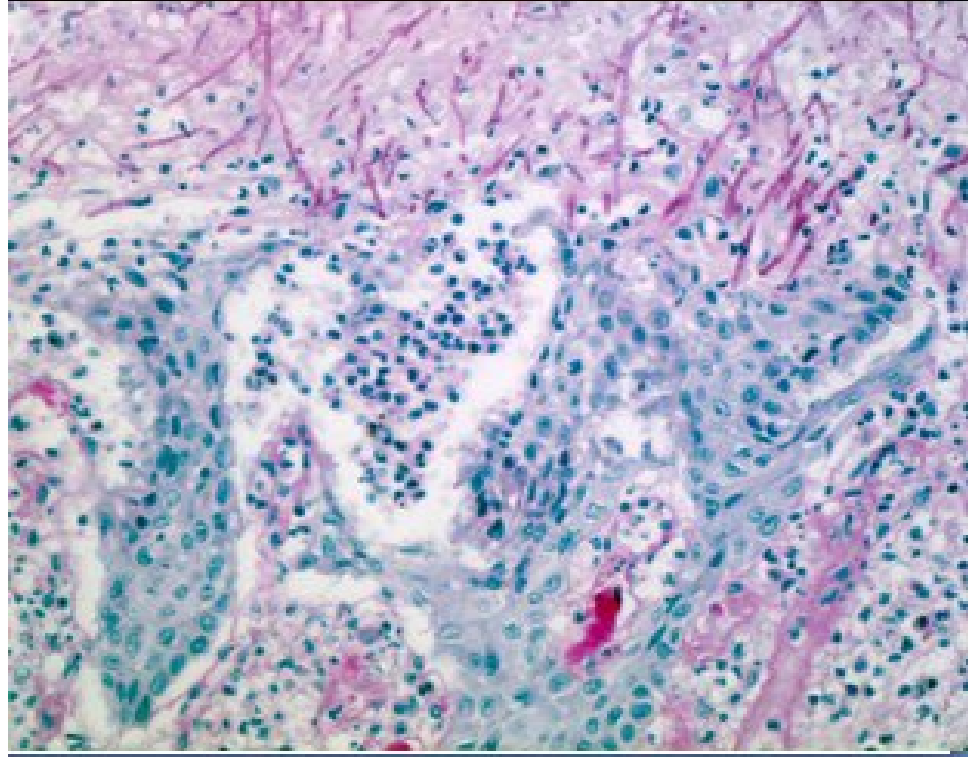
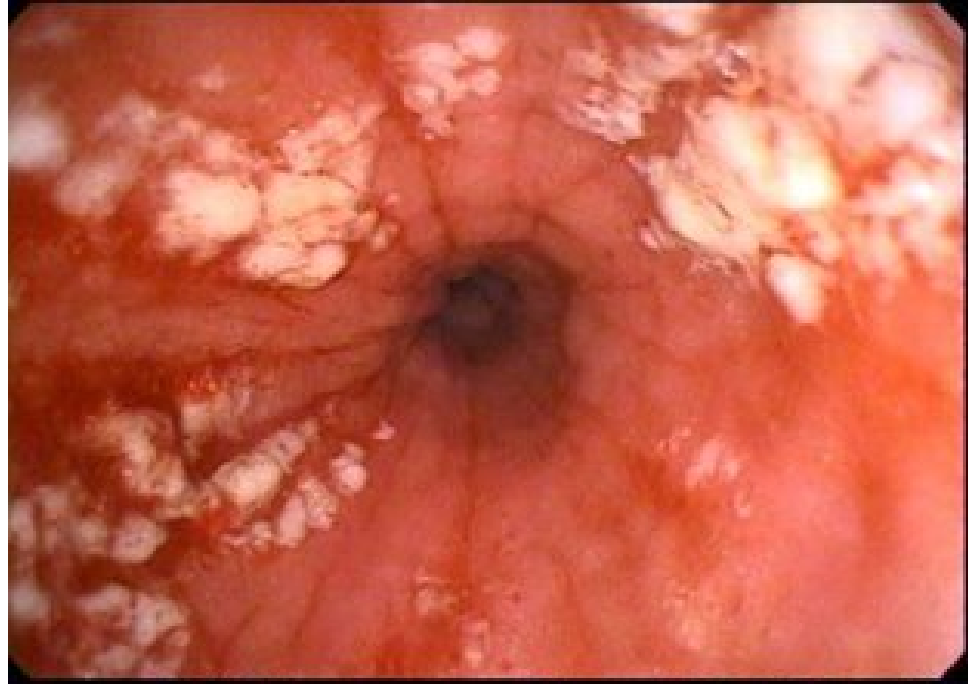
Herpes Esophagitis

- Shallow vesicles and ulcers; may coalesce into extensive areas.
- Multinucleate squamous cells containing herpesvirus nuclear inclusions.



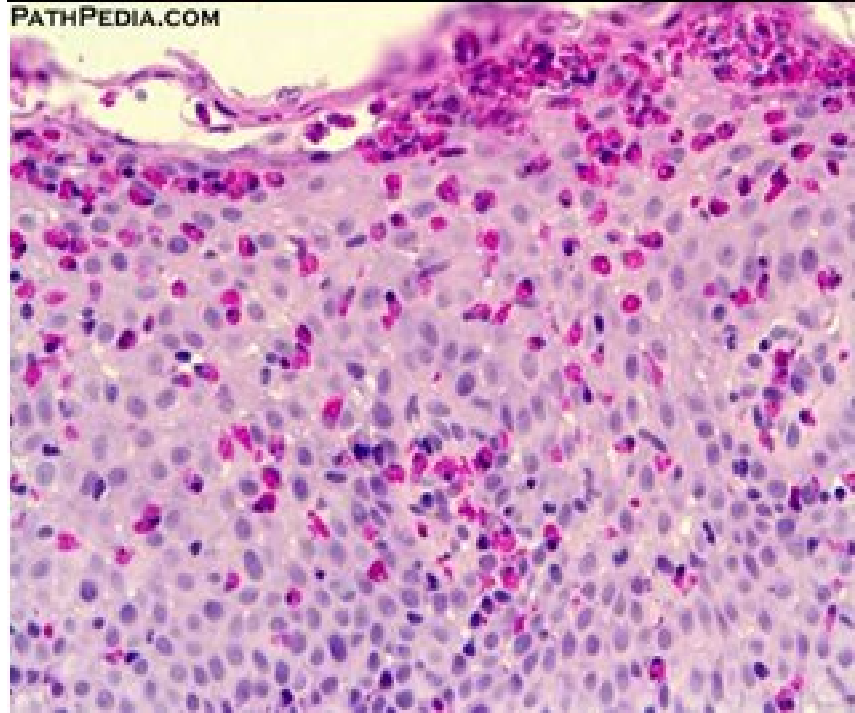
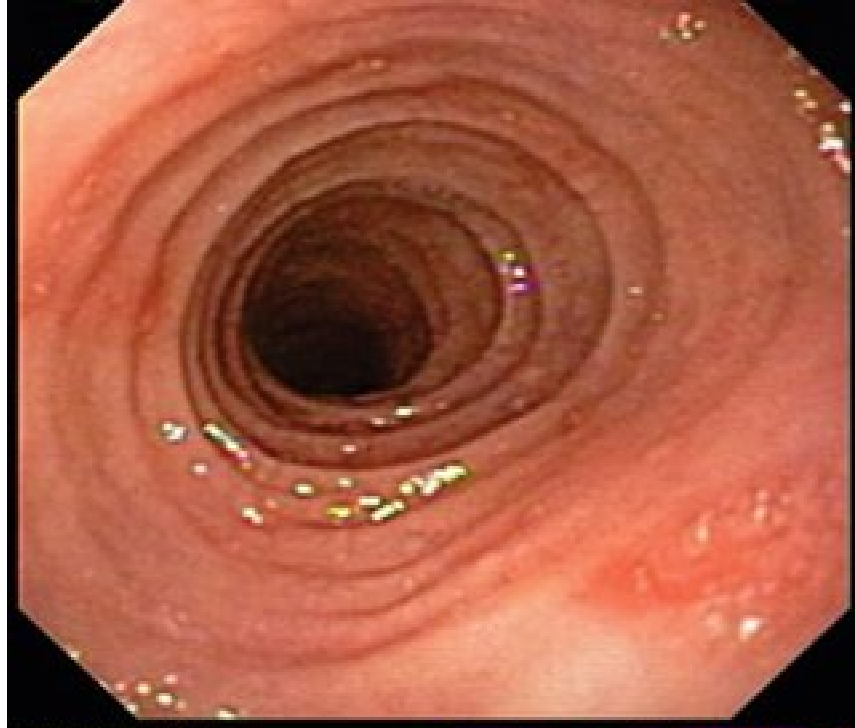
Candidal Esophagitis

- Characterized by adherent, gray white pseudo membranes composed of densely matted fungal hyphae and inflammatory cells covering the esophageal mucosa.



Eosinophilic Esophagitis

- Allergic in etiology
- Patient may have other atopic diseases or peripheral eosinophilia.
- Endoscopically: characteristic transverse ridges which resemble the trachea.
- Histologically: large numbers of eosinophils infiltrate the superficial epithelium and at sites **far** from the GEJ.



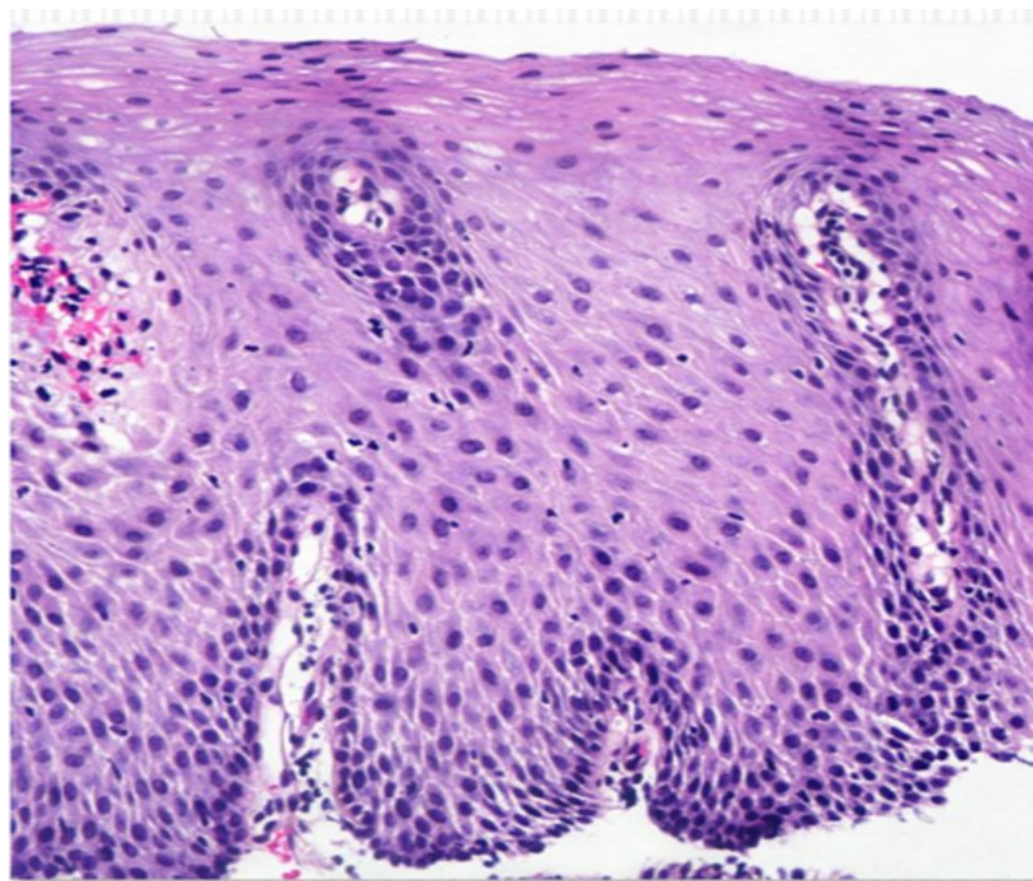
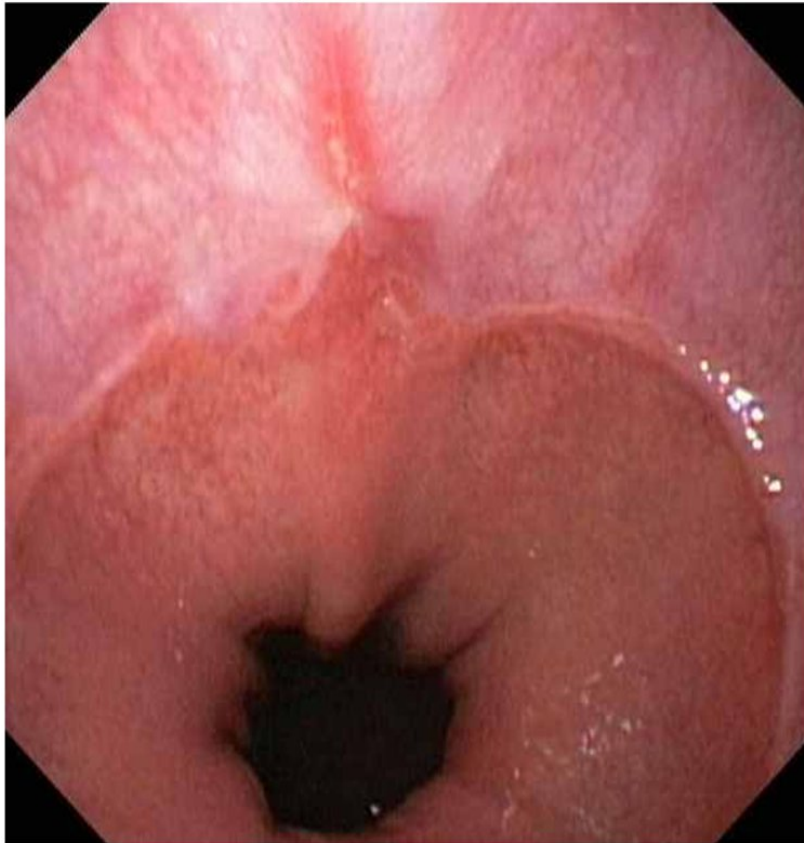
Reflux Esophagitis

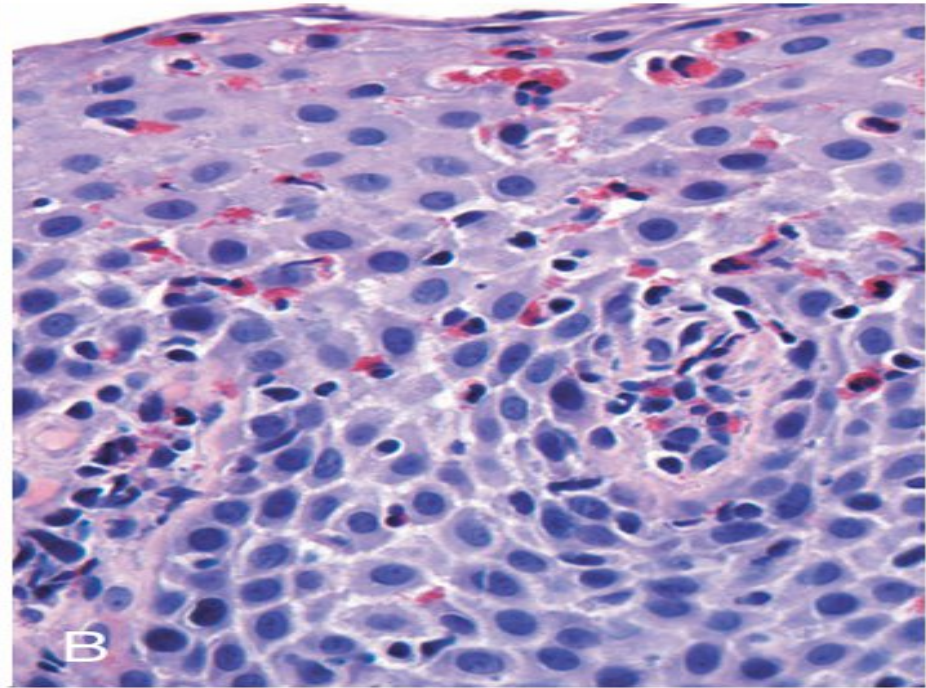
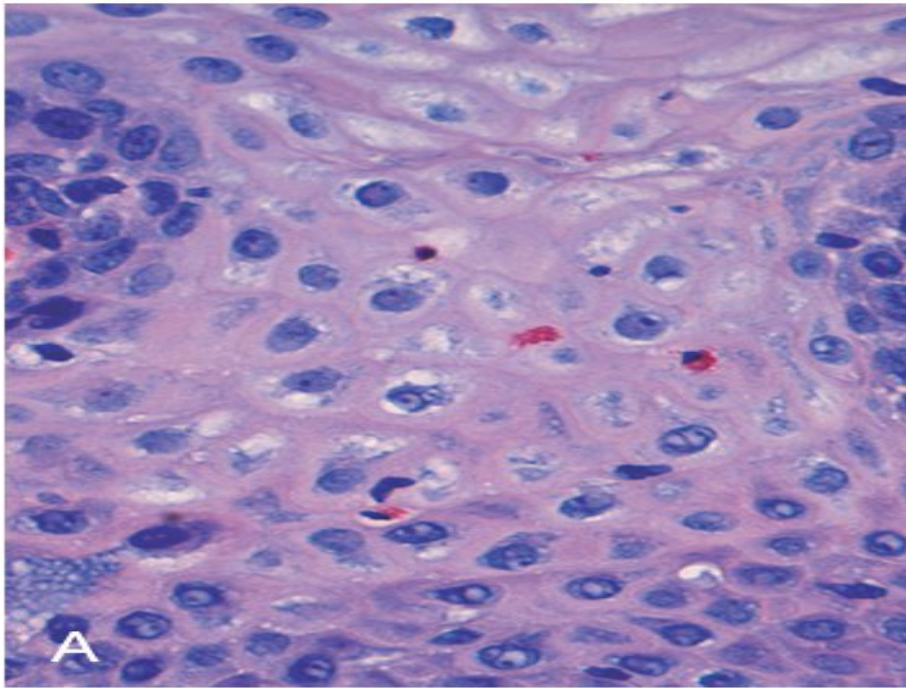
- Inflammation of lower esophageal mucosa caused by reflux of gastric juice.
- Clinically it is known as GERD (Gastroesophageal reflux disease).
- Most common in adults older than 40 years but also occurs in infants and children.
- Symptoms are heartburn, dysphagia, regurgitation of sour-tasting gastric contents.
- Rarely, chronic GERD is punctuated by attacks of severe chest pain that may be mistaken for heart disease.

- Caused by conditions that decrease LES tone &/or increase abdominal pressure:-
 - Alcohol and tobacco use
 - Obesity
 - CNS antidepressants
 - Pregnancy
 - Hiatal hernia
 - Delayed gastric emptying
 - Increased gastric volume.
 - In many cases no definitive cause is identified.

■ Morphology:

- Endoscopy: hyperemia & redness.
- Histology:
 - Damage to superficial squamous epithelium.
 - Eosinophils and then neutrophils infiltrate the epithelium.
 - Basal zone hyperplasia & elongation of lamina propria papillae.





A, Reflux esophagitis with **scattered** intraepithelial eosinophils. **B**, Eosinophilic esophagitis with **numerous** intraepithelial eosinophils.

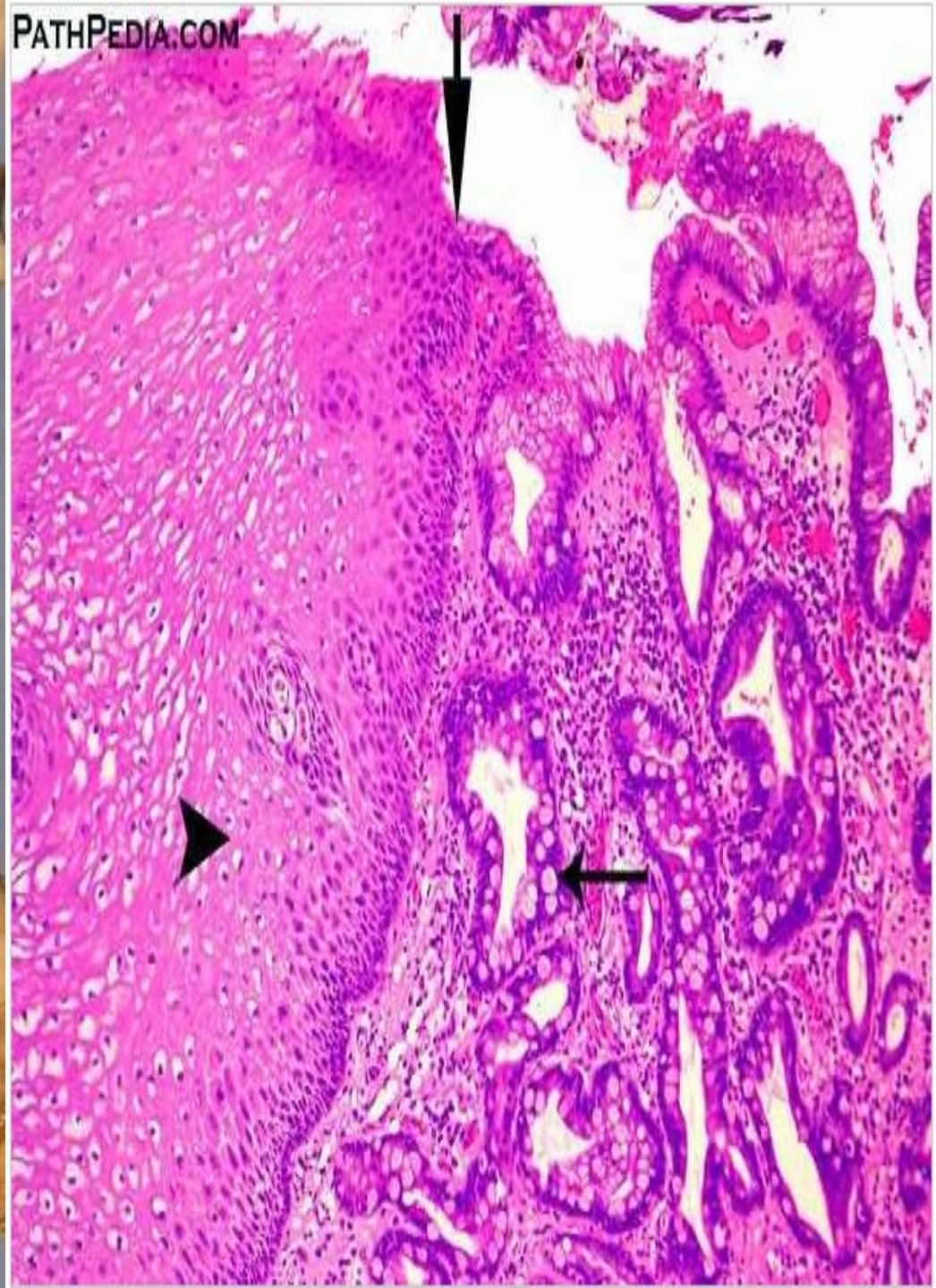
■ **Complications:**

- Esophageal ulceration.
- Hematemesis & melena.
- Stricture development.
- Barrett esophagus.

Barrett Esophagus (BE):

- Acquired glandular metaplasia of esophagus.
- Caused by reflux of gastroduodenal contents in long standing GERD.
- There is a risk of progression to adenocarcinoma.
- Dysplasia could be seen as an intermediate stage between metaplasia and malignancy.

Barrett Esophagus

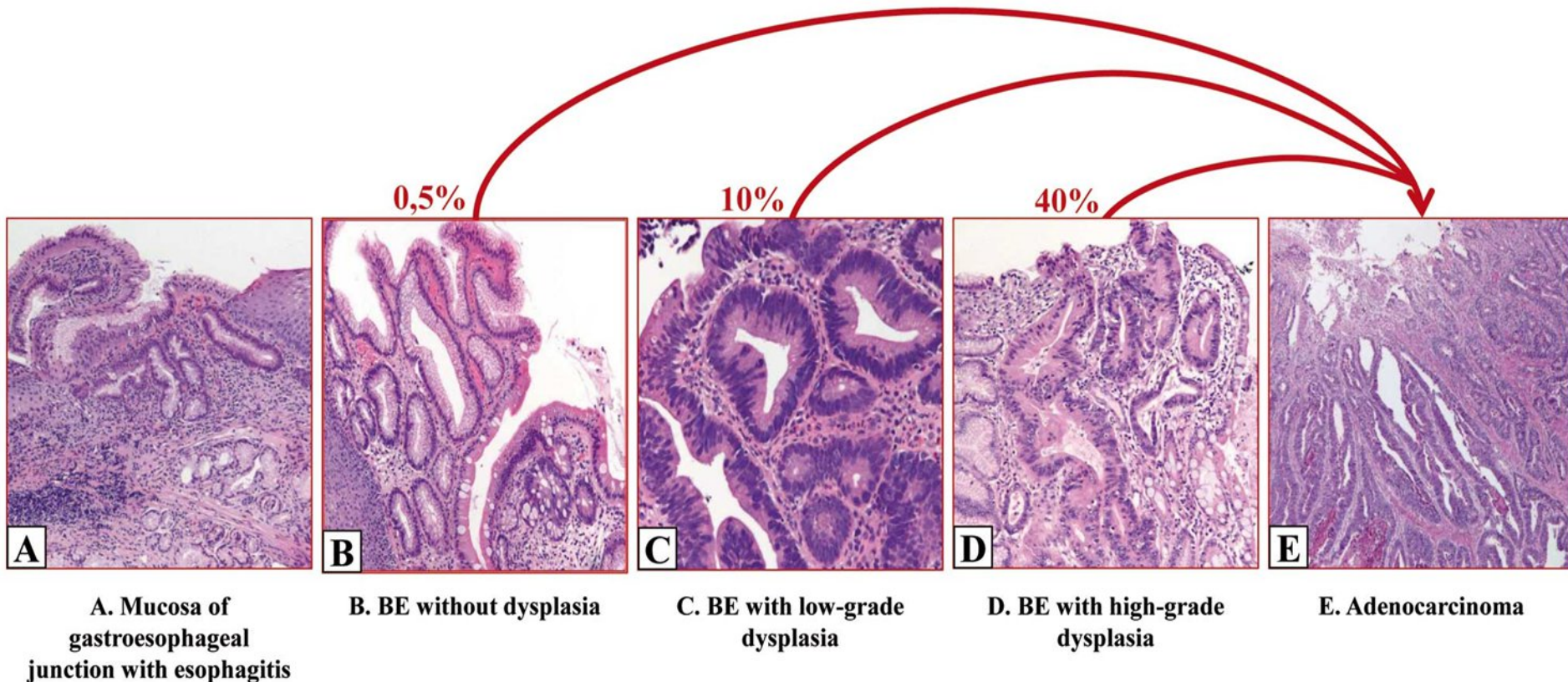


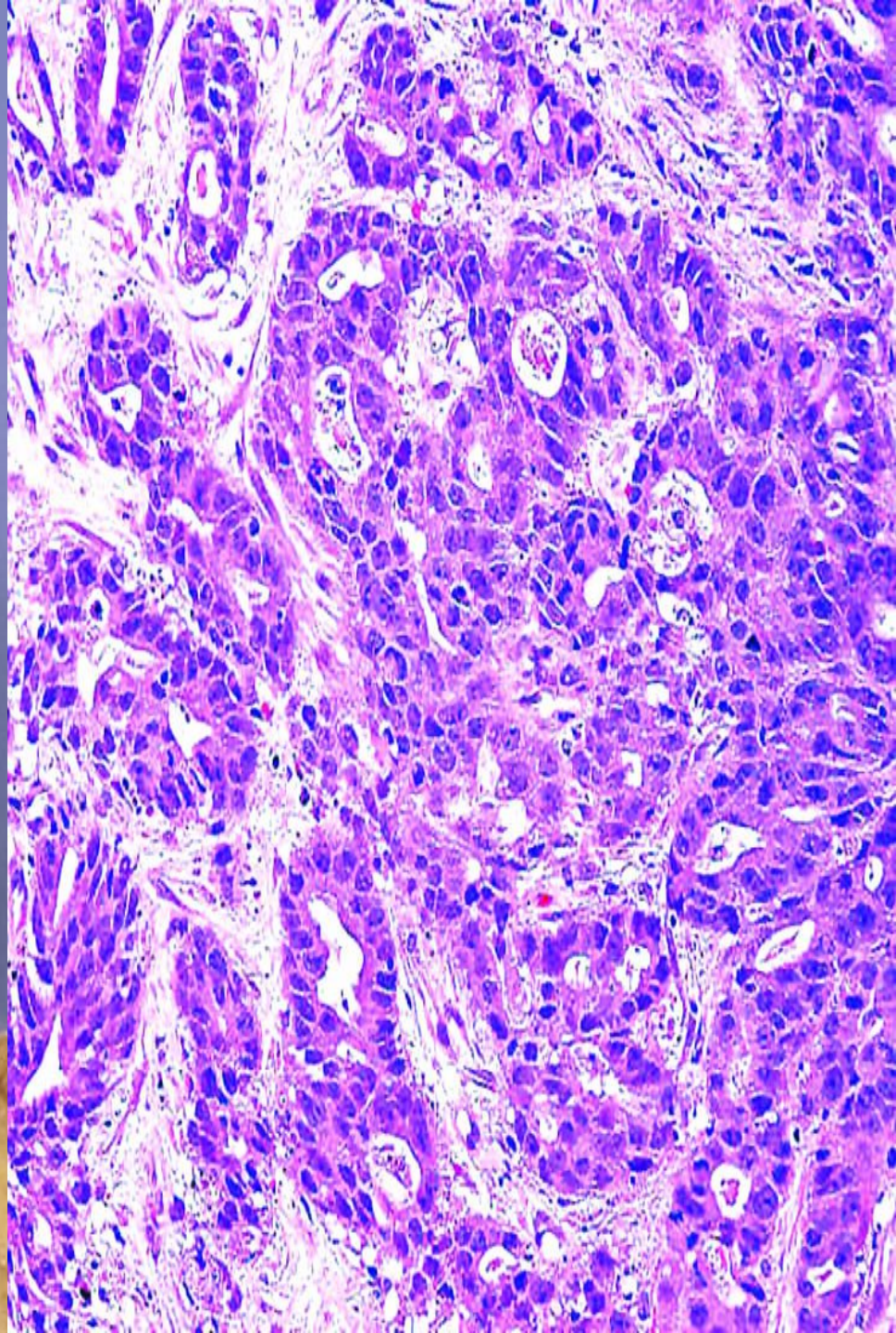
Esophageal tumors:

- **Benign neoplasms:** may arise from:
 - Epithelium: e.g. papilloma
 - Mesenchymal: e.g. leiomyoma which is the commonest.
- **Malignant neoplasms:** Almost all arise from the epithelium; the vast majority are:
 - Adenocarcinoma.
 - Squamous cell carcinoma (SCC).

■ Esophageal adenocarcinoma:

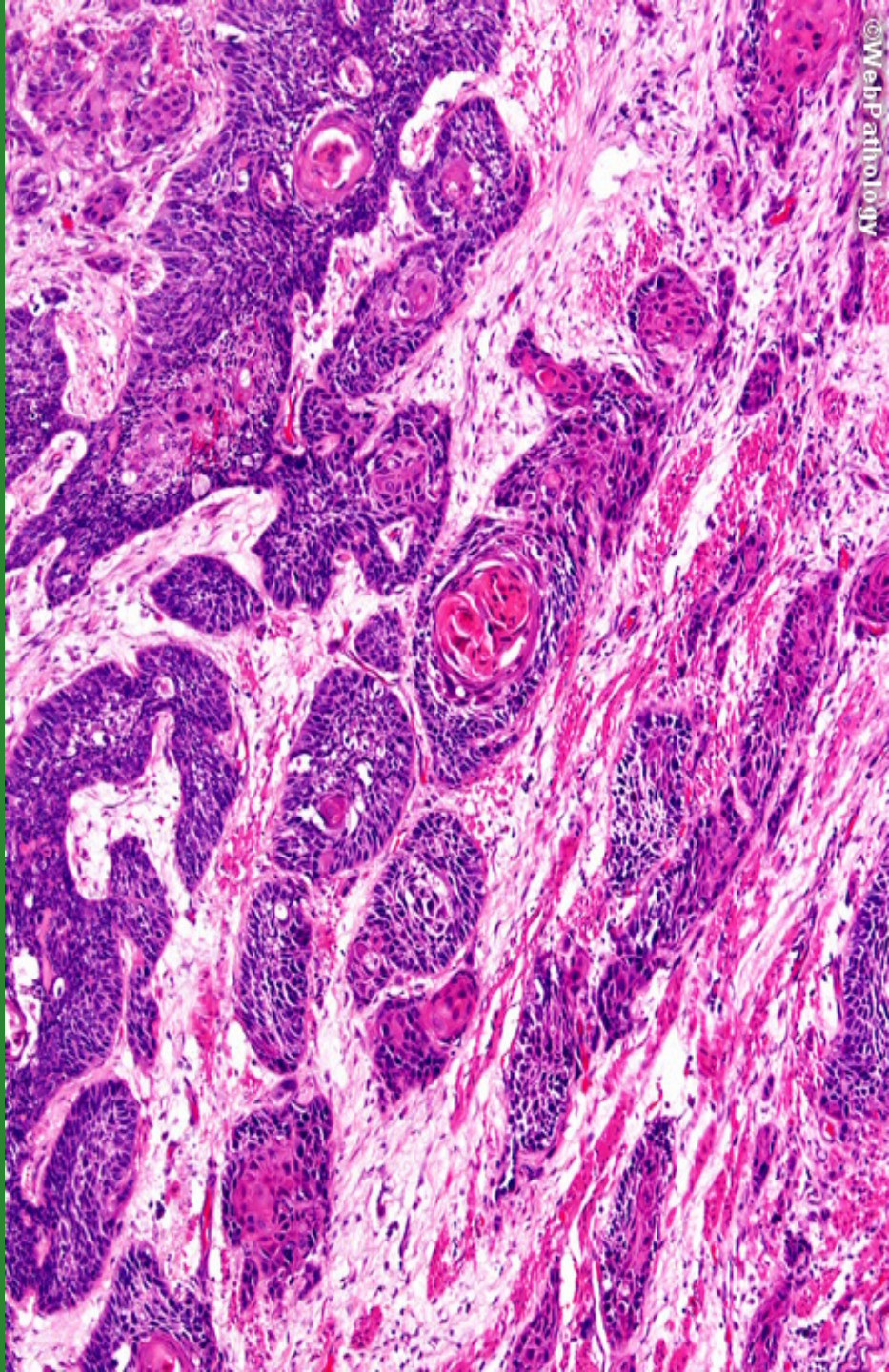
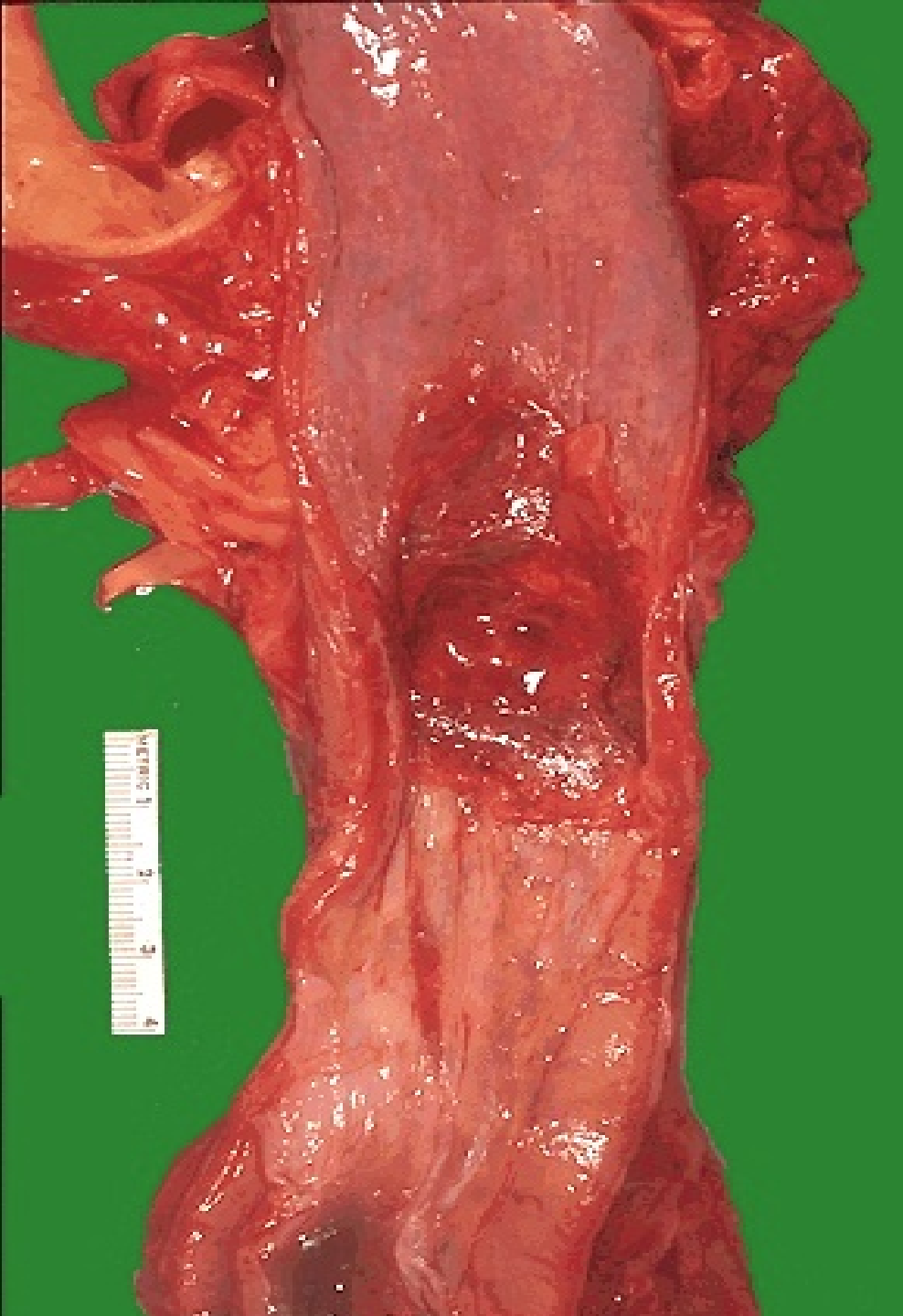
- Most arise from Barrett Esophagus (BE).
- The incidence is increasing worldwide.
- Usually occurs in the distal third of esophagus and may invade the adjacent gastric cardia.





■ **Esophageal SCC:**

- **More common worldwide.**
- **Half of SCC occur in the middle third of esophagus.**
- **Risk factors include:-**
 - **Alcohol and tobacco use.**
 - **Poverty.**
 - **Caustic esophageal injury.**
 - **Achalasia.**
 - **Plummer- Vinson Syndrome.**
 - **Frequent consumption of very hot drinks.**
 - **Previous radiation therapy to mediastinum.**
 - **Nutritional deficiencies; as well as polycyclic hydrocarbons, nitrosamines, and other mutagenic compounds such as those found in fungus contaminated foods.**
 - **HPV infection in high-risk regions.**



Esophageal carcinoma / clinical:

- Usually present in advanced stage.
- Dysphagia is the commonest symptom.
- Odynophagia (pain on swallowing).
- Weight loss and debilitation.
- Hemorrhage and sepsis.
- Symptoms of iron deficiency.
- Aspiration of food via a tracheoesophageal fistula.

- The rich submucosal lymphatic network promotes circumferential and longitudinal spread.
- The sites of LN metastases depend on tumor location:-
 - Upper third: cervical LN.
 - Middle third: mediastinal, paratracheal, and tracheobronchial LN.
 - Lower third: gastric and celiac LN.